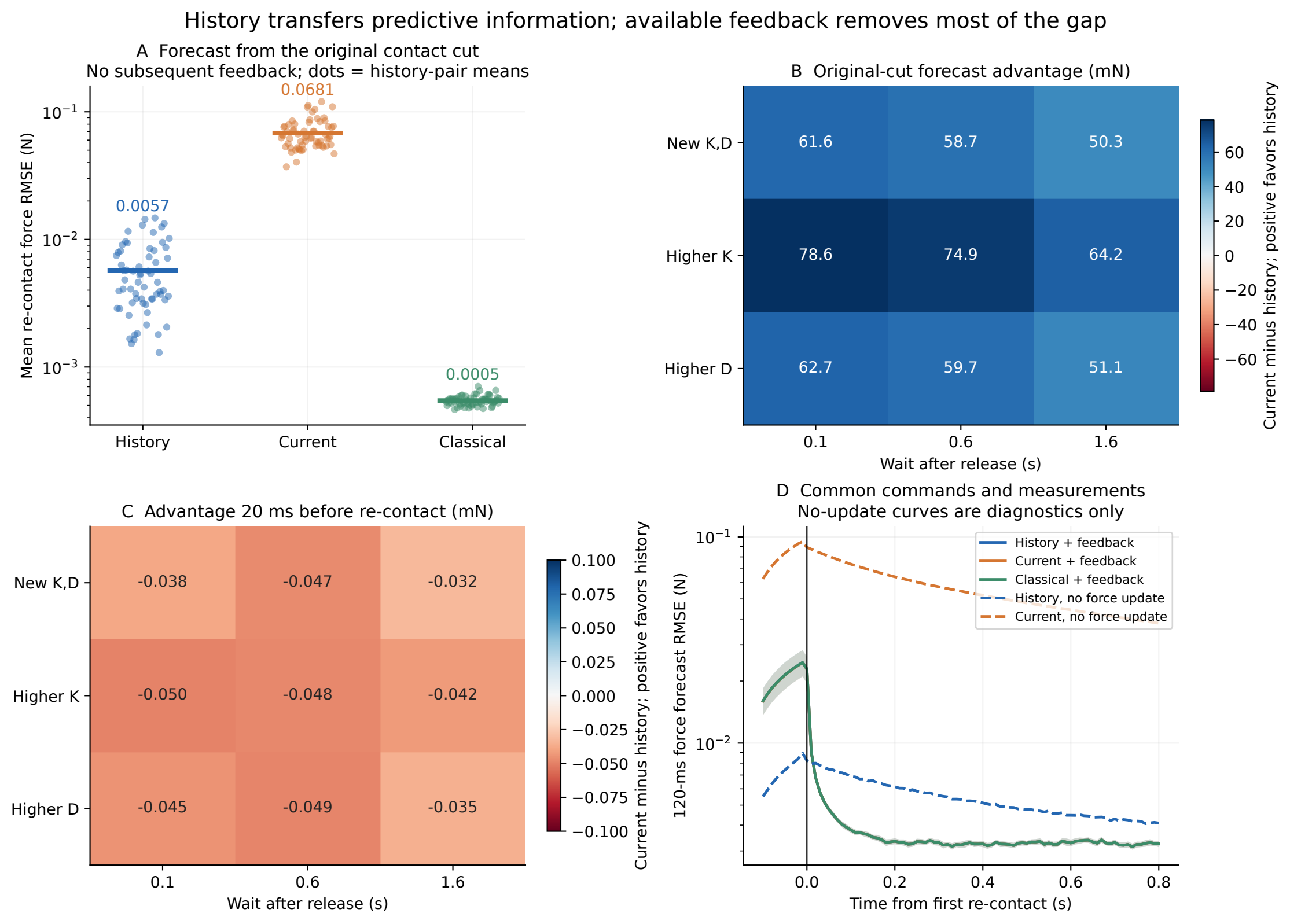

History transfers predictive information; available feedback removes most of the gap
A Forecast from the original contact cut
No subsequent feedback; dots = history-pair means
Mean re-contact force RMSE (N)
0.0681
0.0057
0.0005
History
Current
Classical
B Original-cut forecast advantage (mN)
New K,D
Higher K
Higher D
61.6
58.7
50.3
78.6
74.9
64.2
62.7
59.7
51.1
Wait after release (s)
Current minus history; positive favors history
C Advantage 20 ms before re-contact (mN)
-0.038
-0.047
-0.032
-0.050
-0.048
-0.042
-0.045
-0.049
-0.035
Wait after release (s)
Current minus history; positive favors history
D Common commands and measurements
No-update curves are diagnostics only
History + feedback
Current + feedback
Classical + feedback
History, no force update
Current, no force update
120-ms force forecast RMSE (N)
Time from first re-contact (s)